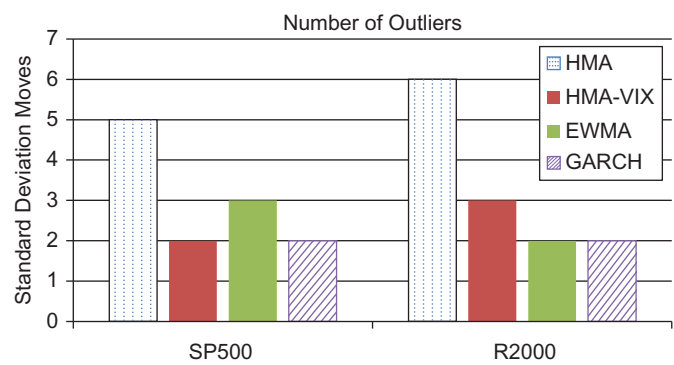


Table 6.3 Performance Results				
Model	SP500		R2000	
	RMSE	RMZSE	RMSE	RMZSE
HMA	0.0106	2.6310	0.0135	4.9373
HMA-VIX	0.0095	2.2965	0.0129	4.1787
EWMA	0.0102	2.1019	0.0131	4.1824
GARCH	0.0103	1.6825	0.0131	3.2585



■ **Figure 6.5** Number of Outliers

fewest outliers with four in total, followed by the HMA-VIX and EWMA with five each. The HMA had the most outliers with 11 in total. The total number of outliers from each model is shown in [Figure 6.5](#).

[Figure 6.6](#) illustrates the ratio of price return divided by predicted HMA-VIX volatility for the SP500 index over our sample period. As shown in this figure, there were two outliers that occurred towards the end of August 2011 during the US debt ceiling crisis.

Under a conservative set of testing criteria, there appears to be compelling statistical evidence that the HMA-VIX adjustment model is as accurate as the EWMA and GARCH models. These are extremely encouraging results since only the HMA-VIX model is capable of reacting in real-time to market events or potential market events that will impact volatility but have not yet affected realized market prices. The historical models will not react to these events until that information is captured in the prices. This